GDES Version 5.0

Comprehensive User Manual & End-to-End Clinical Workflow Guide

Glomerular Disease Expert System

BIRDEM Nephrology · Dhaka, Bangladesh

What's New in GDES Version 5.1

GDES 5.1 delivers a **fully reliable Clinical Decision Support (CDS) layer** — all P0-critical bugs resolved, field-name schema drifts corrected, and the entire CDS pipeline validated through a new integration test suite.

Key Improvements

Clinical Decision Support Now Reliable

The three CDS engines — drug toxicity monitoring, treatment failure detection, and relapse detection — are now fully operational. Previously, these modules could silently fail or produce incorrect results due to:

- **Wrong field names:** `numeric\_value` → `value\_numeric` (the correct field) across 7 query locations
- **Wrong medication source:** The toxicity engine now reads from `TreatmentExposure` (research-grade medication episodes) instead of the broken `Prescription` model query
- **Inverted alert logic:** Low-value-is-worse tests (WBC for MMF/cyclophosphamide, IgG for rituximab) now use `≤` thresholds instead of `≥`, eliminating false negatives for cytopenias and hypogammaglobulinemia
- **Missing disease source:** The diagnosis passed to CDS now reads from the `differential[0].disease\_id` field instead of the non-existent `disease\_trajectory.get("disease\_id")`

Clinicians can now trust the toxicity, failure, and relapse alerts presented on the patient management page.

Human-Readable Disease Names

The differential diagnosis view now shows proper disease names (e.g. "IgA Nephropathy / IgA Vasculitis Nephritis") instead of raw rule IDs (e.g. "TEST-IGA-001"). This fix applies across all CDS output — no more confusing internal identifiers in clinical context.

Cleaner UI — No More Zero-Confidence Clutter

Diagnoses with a 0.0 confidence score are now hidden from the patient detail page. Only rules with meaningful evidence are displayed, reducing visual noise.

Error Transparency

CDS failures are no longer silently swallowed. Each CDS generator block logs exceptions to the Django error log with full tracebacks, and errors are surfaced in a `cds\_errors` list passed to the template context — enabling the UI to show meaningful error states instead of blank panels.

Integration Test Validation

A new `test\_cds\_integration.py` suite (10 tests) validates every CDS pathway against real ORM data — no mocks. The full test suite runs **205 tests passing** with zero regressions.

Summary for Clinicians

The CDS tools you see on the patient detail page — Management Plan, Monitoring Plan, and Follow-up Plan — now produce **accurate, reliable output** backed by automated test validation. If a CDS module encounters an error, it is logged for rapid investigation rather than silently dropped.

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Chapter 1 – Introduction

What is GDES?

The **Glomerular Disease Expert System (GDES)** is a comprehensive clinical information platform purpose-built for the management of glomerular diseases. Developed at BIRDEM Nephrology in Dhaka, Bangladesh, GDES integrates patient registry, clinical decision support, automated follow-up, and research capabilities into a single, unified workflow.

GDES is not merely a database where patient information is stored. It is an **intelligent clinical platform** that:

- Guides clinicians through structured data collection
- Provides real-time diagnostic decision support
- Automatically generates follow-up plans based on disease and risk
- Detects clinical deterioration and escalates care
- Continuously updates research-quality datasets without duplicate data entry

Purpose of the Platform

The primary purpose of GDES is to improve clinical outcomes for patients with glomerular disease through:

1. **Structured, guideline-driven care** — ensuring every patient receives consistent, evidence-based management
2. **Automated clinical reasoning** — helping clinicians recognise disease patterns and avoid diagnostic delays
3. **Systematic follow-up** — reducing loss to follow-up through automated scheduling and escalation
4. **Seamless research integration** — enabling high-quality observational research without additional data collection burden

Target Users

User Role	Primary Activities
Nephrologist	Clinical assessment, treatment decisions, interpretation of clinical reasoning output
Fellow / Resident	Daily patient management, data entry, follow-up visits
Medical Officer	Initial assessment, laboratory entry, prescription generation
Research Coordinator	Study enrolment, data quality monitoring, research dataset export
Nurse	Vital signs, medication administration, patient communication
Data Manager	Data quality, audit trail review, system administration
Hospital Administrator	Dashboard oversight, compliance monitoring, reporting

Clinical Objectives

- Standardise the assessment and management of glomerular diseases
- Provide real-time decision support at the point of care
- Ensure timely follow-up through automated scheduling
- Detect disease progression and relapse early
- Maintain a complete, auditable clinical record for every patient

Research Objectives

- Generate research-quality datasets as a byproduct of clinical care
- Enable survival analysis, Cox regression, and longitudinal outcome studies
- Support registry-embedded clinical trials with randomisation and eligibility screening
- Provide publication-ready data exports for multicentre collaboration

How GDES Differs from a Traditional Registry

Feature	Traditional Registry	GDES
Data entry	Separate research entry	Captured during clinical care
Decision support	None	Real-time reasoning engine
Follow-up	Manual tracking	Automated scheduling with escalation
Prescriptions	Not included	Integrated bilingual prescription printing
Audit trail	Limited	Per-field change history
Trial platform	Separate system	Built-in randomisation and eligibility
Knowledge integration	None	Knowledge base with active rules

GDES combines five capabilities into a single platform:

```
┌─────────────────────────────────────────────────────┐
│                    GDES Platform                     │
│  ┌──────────┐  ┌──────────┐  ┌──────────────────┐  │
│  │ Clinical  │  │ Decision │  │    Knowledge      │  │
│  │ Registry  │  │ Support  │  │     Engine        │  │
│  ├──────────┤  ├──────────┤  ├──────────────────┤  │
│  │ Patient   │  │ Clinical │  │ Rule matching     │  │
│  │ data      │  │ reasoning│  │ Differential      │  │
│  │ Encounters│  │ Risk     │  │ Treatment recs    │  │
│  │ Labs      │  │ assessment│ │ Information gaps  │  │
│  │ Pathology │  │ Care     │  │                   │  │
│  │           │  │ pathway  │  │                   │  │
│  ├──────────┤  ├──────────┤  ├──────────────────┤  │
│  │ Automated │  │ Research  │  │    Prescription    │  │
│  │ Follow-up │  │ Platform │  │     Engine        │  │
│  ├──────────┤  ├──────────┤  ├──────────────────┤  │
│  │ Scheduling│  │ Analytics│  │ Reconciliation    │  │
```

```
│   │ Escalation│   │ Exports   │   │ Safety checks     │   │
│   │ Worklist  │   │ Trials    │   │ Bilingual PDF     │   │
│   └───────────┘   └───────────┘   └───────────────────┘   │
└──────────────────────────────────────────────────────────┘
```

Chapter 2 – System Overview

Dashboard

The dashboard is the home screen of GDES. It provides:

- **Registry at a glance** — total patients, active patients, recent registrations, pending baseline assessments
- **Follow-up Engine** — pending tasks, overdue items, high-priority alerts
- **Follow-up worklist** — visits due now and overdue, updated automatically every 60 seconds
- **Cohort kidney trend** — mean eGFR and proteinuria by month across the entire registry
- **Quick actions** — register a patient, open analytics, export data

> **Clinical Tip:** Check the Follow-up worklist at the start of each clinic day. It shows every patient who is due or overdue for a visit, prioritised by urgency.

Patient Registry

The patient registry is the central hub of GDES. Every patient has a unique Study ID and a permanent record that accumulates all clinical data over time.

Key features:
- Patient search by ID, name, or phone number
- Registration with demographics, comorbidities, and consent
- Automatic calculation of age, CKD-EPI eGFR, and KDIGO stage
- Visual workflow progress strip showing completion status

Automatic processes on registration:
- Audit log entry created
- Research eligibility checked
- Timeline entry generated
- Follow-up engine initialised

Clinical Encounters

Every patient contact is recorded as a Clinical Encounter. This is the central data structure around which all other modules are organised.

Types of encounters:
- Initial consultation
- Follow-up visit
- Unscheduled visit (for acute issues)
- Telephone consultation

Each encounter captures:
- Date and type

- Blood pressure, weight, edema grade
- Clinician response assessment (complete remission, partial remission, stable disease, no response)
- Disease phase (active disease, remission, relapse, etc.)
- Next scheduled visit date

Automatic processes:
- eGFR and proteinuria trend chart updated
- Disease phase may be automatically updated
- Clinical reasoning triggered
- Follow-up plan reassessed

Clinical Assessment

The baseline assessment is completed at enrolment and captures:
- Chief complaint and presenting symptoms
- Syndrome classification (nephrotic syndrome, nephritic syndrome, rapidly progressive GN, etc.)
- Full vital signs and anthropometrics
- Comorbidity assessment

Automatic calculations:
- BMI (with Asian-specific categories)
- Body surface area (BSA)
- CKD-EPI 2021 eGFR
- KDIGO stage (G1–G5)
- Proteinuria category

Laboratory

GDES captures all laboratory data relevant to glomerular disease management.

Key tests:
- Creatinine and eGFR
- Albumin
- 24-hour urine protein (UTP)
- Urine protein-to-creatinine ratio (UPCR)
- Urine albumin-to-creatinine ratio (UACR)
- Complete blood count (CBC)
- Complement C3, C4
- Autoantibodies (ANA, anti-dsDNA, ANCA, anti-PLA2R)
- Gd-IgA1
- HbA1c (for diabetic patients)

Automatic actions:
- Reference range checking with critical value alerts
- Longitudinal trend chart updated
- eGFR slope computed after sufficient data points
- AKI detection (KDIGO criteria)
- CKD progression detection

- Timeline updated
- Research dataset synchronised

> **Clinical Tip:** Trends matter more than single values. The lab trend chart on the patient overview shows eGFR and proteinuria together, making it easy to assess disease trajectory at a glance.

Pathology

GDES supports structured entry of kidney biopsy pathology.

Key features:
- Biopsy date, laterality, adequacy assessment
- Light microscopy, immunofluorescence, and electron microscopy findings
- Disease-specific scoring systems:
- Oxford MEST-C for IgA Nephropathy
- ISN/RPS classification for Lupus Nephritis
- Columbia classification for FSGS
- PLA2R staining for Membranous Nephropathy
- Central review workflow with concordance tracking

Automatic actions on biopsy entry:
- Disease classification assigned
- Knowledge engine activated with disease-specific rules
- Clinical reasoning triggered
- Timeline updated
- Research dataset updated

Clinical Reasoning

The Clinical Reasoning module is one of GDES's most powerful features. It automatically analyses every patient's complete data profile and generates:

- **Differential diagnosis** — ranked list of possible diagnoses with confidence scores
- **Risk assessment** — progression risk, relapse risk, kidney survival estimate
- **Care pathway** — current disease stage, treatment recommendations, care gaps
- **Information gaps** — missing data that would improve diagnostic accuracy

The reasoning engine is rule-based, drawing on a curated knowledge base of glomerular disease patterns. Each rule is evidence-graded and linked to guideline references.

> **Clinical Tip:** The Clinical Reasoning tab is available for every registered patient. Check it after each data entry to see if new information has changed the differential or risk assessment.

Decision Support

The Decision Support module extends Clinical Reasoning by providing:

- Evidence grades for each recommendation
- Specific guideline references (KDIGO, ISN, etc.)

• Treatment suggestions linked to disease type and phase
• Drug monitoring requirements based on active therapy
• Drug-drug interaction warnings
• Contraindication alerts

How it works:
1. Patient data is analysed
2. Knowledge base rules are matched
3. Recommendations are generated with confidence levels
4. Clinician reviews and accepts, modifies, or overrides
5. Overrides are recorded in the audit log with reason

Treatment

GDES tracks treatment through two complementary systems:

Treatment Exposures — Research-grade medication episodes with start/stop dates, dose, regimen, and stop reason. These are automatically reconciled whenever a prescription is finalised.

Prescriptions — Clinical prescription documents that are printed for the patient. Each prescription can have multiple items with dose, frequency, and duration.

Key features:
- Drug selection from a curated formulary
- Dose calculation assistance
- Contraindication checking
- Drug interaction warnings
- Monitoring requirements display

Automatic reconciliation on prescription finalisation:
- New drug without existing exposure → episode opened
- Drug dropped from regimen → episode closed with stop reason
- Dose or regimen changed → old episode closed, new one opened (episode split)
- Unchanged drug → episode continued

Prescriptions

GDES generates bilingual (English and Bengali) prescription PDFs suitable for printing and direct patient use.

Prescription features:
- Versioned (each modification creates a new version)
- Immutable once finalised (medico-legal)
- Cryptographic hash for tamper detection
- Drug name, dose, frequency, duration, and indications
- Clinician name and signature line
- Hospital header and patient identification

Workflow:
1. Create prescription with items

2. Review and modify as needed
3. Finalise (makes immutable and triggers reconciliation)
4. Preview on screen
5. Download PDF for printing

> **Clinical Tip:** The prescription *is* the follow-up visit record. Finalising a prescription simultaneously generates the visit note, updates the medication list, and triggers the follow-up engine.

Follow-up

The Follow-up module is an automated engine that manages every patient's long-term care schedule.

Key capabilities:
- Disease-specific follow-up protocols (IgAN, MCD, FSGS, MN, LN, ANCA vasculitis, anti-GBM, C3G, MPGN, DDD)
- Risk-stratified follow-up intervals (high risk → shorter interval)
- Automated task generation (visit due, lab due, medication monitoring, safety monitoring, research visit, referral due, patient contact)
- Multi-level escalation for overdue tasks
- Clinician worklist dashboard

For a detailed explanation, see Automated Follow-up Engine.

Timeline

Every patient has a chronological timeline of all significant events:

- Registration and consent
- Baseline assessment
- Follow-up visits
- Laboratory results
- Biopsies
- Hospital admissions
- Relapses
- Medication changes
- Outcome computations

The timeline is automatically populated and updated — no manual entry required.

Knowledge Base

The Knowledge Base is the curated repository of glomerular disease knowledge that powers the Clinical Reasoning engine.

Contents:
- Disease definitions and diagnostic criteria
- Clinical feature-disease associations

- Diagnostic rules with evidence grading
- Treatment guidelines
- Follow-up protocol definitions

The Knowledge Base is accessible via the admin interface and can be extended by authorised users as new evidence emerges.

Research

GDES treats research as a byproduct of clinical care. Every data point entered during routine clinical work is automatically available for research.

Research capabilities:
- **Outcome computation** — eGFR slope, remission status, time to remission, ESKD, death, composite endpoints
- **Survival analysis** — Kaplan-Meier curves with Greenwood confidence intervals, log-rank testing, incidence rates
- **Cox proportional hazards** — multivariable regression
- **Competing risks analysis** — cumulative incidence functions
- **eGFR slope analysis** — mixed-effects models
- **Research dataset export** — one-row-per-patient denormalised dataset in CSV or Excel
- **Registry-embedded trials** — study definition, arm randomisation, eligibility screening, DSMB reporting

> **Clinical Tip:** To compute outcomes for a patient, open their record and click "Compute now" on the Outcomes card. This triggers the full outcome engine and updates the research dataset.

Administration

The Administration section provides:

- User management (invite, roles, permissions)
- Audit log review
- System configuration
- Backup management
- Knowledge base management (for authorised users)

Chapter 3 – Complete Patient Journey

This chapter follows a realistic patient through the entire GDES workflow from first presentation to long-term follow-up.

Case Introduction

Mr. Rahman is a 35-year-old male who presents to the BIRDEM Nephrology clinic with:

- Progressive bilateral lower limb edema over 3 weeks
- Fatigue and reduced exercise tolerance
- Foamy urine noticed by the patient
- Blood pressure of 148/92 mmHg at presentation
- No prior kidney disease
- No diabetes, no hypertension history
- No family history of kidney disease

Initial investigations ordered by the referring physician show:
- Serum creatinine: 1.6 mg/dL (eGFR 48 mL/min/1.73m²)
- Serum albumin: 2.8 g/dL
- 24-hour urine protein: 3.8 g/day
- Urinalysis: 2+ protein, 3+ blood, dysmorphic RBCs
- HbA1c: 5.2% (non-diabetic)

Step 1 – Patient Registration

Creating the Patient Record

1. From the Dashboard, click **"Register New Patient"** in the task cards.
2. The registration form opens with the following fields:

Demographics Section:
- Full name: *Md. Rahman*
- Age / Date of birth: *15-Jan-1991*
- Sex: *Male*
- Phone number: *017XXXXXXXX*
- Address: *Dhaka, Bangladesh*
- Hospital registration number (if available)

Medical History Section:
- Diabetes status: *None*
- Hypertension status: *Yes (newly diagnosed)*

- Prior kidney disease: *No*
- Family history of kidney disease: *No*

1. Click **Save** to create the record.

Automatic Actions on Registration

Once the patient is saved, GDES performs several actions automatically:

```
Registration Saved
│
├─► Audit Log: "Patient registered by Dr. Khan"
│
├─► Research Eligibility: Checked against active studies
│
├─► Timeline: "Patient registered" entry created
│
├─► Follow-up Engine: Initialised (default protocol)
│
├─► Clinical Reasoning: Pending (waiting for baseline data)
│
└─► Patient ID: A1-001 assigned
```

Consent

After registration, navigate to the **Consent** tab to manage consent:

1. Click **"Manage"** in the Consent tab.
2. Review each consent type:
3. Registry participation
4. Biobank sample storage
5. Study enrolment (separate for each study)
6. For each, select **"Granted"** and enter the consent date.
7. Click **Save**.

> **Clinical Tip:** Consent can be withdrawn at any time. Withdrawn consents are recorded in the audit log. For biobank consent, withdrawal means existing samples are retained but no new samples are collected.

Patient Record After Registration

After registration, the patient detail page shows:

- **Workflow progress:** Registration ☑ complete
- **Status badge:** "Registered"
- **Summary cards:** Demographics, baseline (pending), outcomes (pending)
- **Tabs:** Overview, Clinical Reasoning, Visits, Prescriptions, Labs, Safety, Biopsies, Research, Consent, Audit Log

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Step 2 – Baseline Assessment

Completing the Baseline Form

From the patient detail page, click **"Baseline"** in the Stage 1 actions bar.

The baseline assessment form captures:

Clinical Examination:
- Date of assessment
- Systolic BP: *148 mmHg*
- Diastolic BP: *92 mmHg*
- Weight: *68 kg*
- Height: *168 cm*
- Edema grade: *2+ (moderate, bilateral lower limb)*
- Chief complaint: *Progressive edema*

Syndrome Classification:
- Presentation syndrome: *Nephrotic syndrome*

Comorbidities:
- Hypertension: *Yes (newly diagnosed)*
- Diabetes: *No*
- Other: *None*

Automatic Calculations

When the form is saved, GDES automatically computes:

Calculation	Result	Interpretation
BMI	24.1 kg/m²	Overweight (Asian criteria)
BSA	1.77 m²	Normal
eGFR (CKD-EPI 2021)	48 mL/min/1.73m²	CKD G3b
KDIGO stage	G3b A3	High risk
Proteinuria category	3.8 g/day	Nephrotic range

What Happens Next

```
Baseline Assessment Saved
|
├─► Audit Log: "Baseline assessment recorded by Dr. Khan"
|
├─► eGFR: Automatically calculated and stored
├─► CKD Stage: G3b assigned
├─► KDIGO Risk: "High risk" assigned
|
├─► Timeline: "Baseline assessment" entry created
|
```

```
├─► Clinical Reasoning: Triggered (partial data available)
├─► Differential: Preliminary evaluation
├─► Information gaps: Identified
│
├─► Follow-up Engine: Risk assessed for interval determination
├─► Follow-up interval: 90 days (high risk, new patient)
│
└─► Research Dataset: Updated with baseline values
```

The workflow progress now shows:
- Registration ☑
- Baseline ☑
- Prescription ✖ pending
- Follow-up ✖ pending
- Outcomes ✖ pending

Step 3 – Laboratory Entry

Laboratory results can be entered in two ways:

1. **At a baseline or follow-up visit** — using the lab section of the visit form
2. **Standalone** — using the "Enter results" link in the Follow-up stage bar

Entering Initial Labs

For Mr. Rahman's initial presentation, the following labs were ordered and are now being entered:

Core labs (at baseline visit):

Test	Result	Unit	Reference Range
Creatinine	1.6	mg/dL	0.7–1.2
eGFR	48	mL/min/1.73m^2	>90
Albumin	2.8	g/dL	3.5–5.0
24h UTP	3.8	g/day	<0.15
UPCR	3.2	g/g	<0.2
Hemoglobin	12.5	g/dL	13–17
Potassium	4.2	mmol/L	3.5–5.0

Autoimmune workup:

Test	Result	Interpretation
C3	85	Normal (80–160 mg/dL)
C4	28	Normal (15–45 mg/dL)
ANA	Negative	—
ANCA	Negative	—
Anti-PLA2R	Pending	—
Gd-IgA1	Pending	—

Automatic Actions on Lab Entry

```
Lab Results Saved
│
├─► Reference Range Check: All values within range
├─► No critical alerts triggered
│
├─► Trend Chart: First data point added
├─► Future trends will show progression
│
├─► eGFR Slope: Insufficient data (only 1 point)
├─► Will compute after 3+ readings
│
├─► Timeline: "Lab results recorded" entry created
│
├─► Clinical Reasoning: Retriggered with lab data
├─► Differential: Narrowing (low C3 diseases ruled out)
│
├─► Research Dataset: Updated with lab values
│
└─► Follow-up Engine: No change to interval yet
```

> **Clinical Tip:** Lab results are automatically plotted in the trend chart. Access it from the Overview tab of any patient. The chart shows eGFR on the left axis and proteinuria on the right axis, making it easy to see both trends simultaneously.

Step 4 – Biopsy

The clinical presentation (nephrotic syndrome with hematuria, reduced eGFR, negative autoimmune serology) strongly suggests primary glomerulonephritis. Mr. Rahman undergoes a kidney biopsy.

Entering Biopsy Results

From the patient detail page, click **"Add biopsy"** in the Biopsies tab.

Biopsy details:
- Biopsy date: *7 days after registration*
- Laterality: *Left kidney*
- Needle passes: *2*
- Cortex contains: *>10 glomeruli (adequate)*
- Biopsy adequacy: *Adequate*

Light Microscopy:
- Glomeruli: *10 of 12 show mesangial hypercellularity*
- Tubules: *Mild tubular atrophy (<25%)*
- Interstitium: *Mild interstitial fibrosis (<25%)*
- Vessels: *Mild arteriosclerosis*

Oxford MEST-C Scoring:

Component	Score	Description
M (Mesangial hypercellularity)	M1	>50% of glomeruli
E (Endocapillary hypercellularity)	E0	Absent
S (Segmental sclerosis)	S0	Absent
T (Tubular atrophy/interstitial fibrosis)	T1	25–50%
C (Crescents)	C0	Absent

Immunofluorescence:
- IgA: *3+ (dominant, mesangial)*
- IgG: *Negative*
- IgM: *Trace*
- C3: *2+*
- C1q: *Negative*

Electron Microscopy:
- Mesangial electron-dense deposits
- No subepithelial or subendothelial deposits

Final Diagnosis:
- Diagnosis: *IgA Nephropathy (Primary)*

Automatic Actions on Biopsy Entry

```
Biopsy Saved
│
├─► Audit Log: "Biopsy recorded by Dr. Khan"
│
├─► Diagnosis: "IgA Nephropathy, Primary" assigned
│
├─► Disease Classification: Updated to IgA Nephropathy
│
├─► Knowledge Engine: Activated with IgAN-specific rules
├─► IgAN protocol selected for follow-up
│
├─► Timeline: "Biopsy result: IgAN (M1 E0 S0 T1 C0)" entry
│
├─► Clinical Reasoning: Full retrigger with biopsy data
├─► Differential: IgAN confirmed
├─► Risk: IgAN-specific risk assessment
│
├─► Follow-up Engine: Protocol switched to IgAN-specific
├─► Follow-up interval: Recalculated per IgAN protocol
│
└─► Research Dataset: Updated with histopathology data
```

Workflow Status After Biopsy

The workflow progress strip now shows:
- Registration ☑
- Baseline ☑
- Prescription ✖ (pending)
- Follow-up ✖ (pending)
- Outcomes ✖ (pending)

The Clinical Reasoning tab is now populated with a full analysis.

Step 5 – Clinical Reasoning

After the biopsy, Mr. Rahman's Clinical Reasoning tab is fully populated. Here is what GDES displays:

Differential Diagnosis

Disease	Score	Confidence	Source
IgA Nephropathy	95.0	98%	Biopsy confirmed
IgA Vasculitis (Henoch-Schönlein)	45.0	40%	No extrarenal features
Thin Basement Membrane Nephropathy	20.0	15%	Not consistent with presentation
Lupus Nephritis	10.0	5%	ANA negative, C3/C4 normal

Risk Assessment

Risk Factor	Value
Progression Risk	35% (5-year)
MEST-C T1 score predicts higher risk	
Relapse Risk	Not applicable (new diagnosis)
Kidney Survival Estimate	85% at 10 years

Information Gaps

- Gd-IgA1 level (not yet resulted)
- Anti-PLA2R (not yet resulted — useful to rule out MN)
- 24-hour BP monitoring (home BP readings)
- Family history of IgAN (specific query)

Care Pathway

- **Current Stage:** Active disease, newly diagnosed
- **Treatment Recommendations:** 2 recommendations available
- **Care Gaps:** 1 (start RAAS inhibition, assess indication for immunosuppression)

Disease Trajectory

- **Trend:** Newly diagnosed (insufficient follow-up data for trajectory)
- **Next assessment:** 90 days

> **Clinical Tip:** The Clinical Reasoning analysis updates automatically every time new data is entered. Check it after each visit to see how the risk assessment and recommendations evolve over time.

Step 6 – Treatment Planning

Reviewing Decision Support Recommendations

Based on Mr. Rahman's clinical profile, GDES generates the following recommendations:

Recommendation 1 — RAAS Inhibition
- Agent: *Ramipril 5 mg daily*
- Evidence: *KDIGO 2021, Grade 1B*
- Rationale: *Proteinuria >1 g/day, hypertension, eGFR <60*
- Monitoring: *Creatinine and potassium at 2 weeks*

Recommendation 2 — Immunosuppression Assessment
- Indication: *T1 score (tubular atrophy >25%) increases risk of progression*
- Current proteinuria: *3.8 g/day despite optimal RAAS blockade*
- Suggestion: *Consider glucocorticoids*
- Evidence: *TESTING trial (reduced risk of kidney failure)*
- Caution: *Discuss risks (infection, diabetes, bone health)*

Entering Medications

Step 1: Click **"Prescription"** in the Stage 3 actions bar.

Step 2: Add prescription items:

Drug	Dose	Frequency	Duration	Indication
Ramipril	5 mg	Once daily	Ongoing	Proteinuria, hypertension
Atorvastatin	20 mg	Once daily	Ongoing	Hyperlipidemia (nephrotic)

Step 3: Click **"Finalise"** to complete the prescription.

Automatic Actions on Prescription Finalisation

```
Prescription Finalised
│
├─► Audit Log: "Prescription v1 finalised by Dr. Khan"
│
├─► Treatment Reconciliation:
├─► Ramipril → New exposure created (start: today)
├─► Atorvastatin → New exposure created (start: today)
```

```
│
├─► Safety Checks:
├─► No drug-drug interactions detected
├─► Monitoring requirements noted
│
├─► Follow-up Engine:
├─► Monitoring tasks created:
│   ├─► Serum creatinine + potassium (2 weeks)
│   └─► Lipid profile (3 months)
├─► Next visit scheduled (90 days)
│
├─► Timeline: "Prescription v1: Ramipril, Atorvastatin" entry
│
└─► Research Dataset: Updated with treatment exposure
```

Prior Medications (Optional)

The **"Prior meds"** button in Stage 3 allows entry of medications the patient was taking before the first GDES prescription. This is important for research-grade exposure history.

For Mr. Rahman, no prior GN-specific medications were taken.

Step 7 – Automatic Follow-up Planning

The Follow-up Engine is one of GDES's most powerful automation features. After Mr. Rahman's data is complete (registration, baseline, labs, biopsy, prescription), the engine generates a comprehensive follow-up plan.

How the Follow-up Plan is Generated

```
Patient Data → Clinical Profile
│
├─► Disease: IgA Nephropathy
├─► Risk Stratification:
│   ├─► eGFR <60 → +1 point
│   ├─► Proteinuria >1 g/day → +2 points
│   ├─► Hypertension → +1 point
│   ├─► MEST-C T1 → +2 points
│   └─► Total: 6 points → HIGH risk
│
├─► Protocol Selection:
│   ├─► Disease: IgAN-specific protocol selected
│   └─► Risk: HIGH → shorter interval
│
├─► Generated Tasks:
│   ├─► Visit due (108 days)
│   ├─► Lab: creatinine, eGFR, UPCR (2 weeks for safety monitoring)
│   ├─► Lab: full panel (108 days)
│   └─► Medication monitoring: creatinine + K+ (2 weeks)
```

```
│
└► Output → Patient Follow-up Plan
```

The Follow-up Plan

Mr. Rahman's generated follow-up plan:

Task	Due Date	Priority	Status
Safety labs (creatinine, K+)	14 days	High	Pending
Follow-up visit	108 days	Medium	Pending
Full lab panel	108 days	Medium	Pending
Lipid profile	3 months	Low	Pending

Viewing the Follow-up Plan

The follow-up plan is visible in two places:

1. **Patient Overview** — A "Follow-up Plan" card shows all tasks in a table
2. **Dashboard** — The Follow-up worklist aggregates all pending tasks across all patients

What If a Visit is Missed?

The escalation engine monitors all tasks. If Mr. Rahman misses his 108-day visit:

Days Overdue	Escalation Level	Action
1–14	Level 1 (Notice)	Task marked overdue, visible in worklist
15–30	Level 2 (Alert)	Notification to clinician dashboard
31–60	Level 3 (Warning)	Escalation to clinic coordinator
>60	Level 4 (Critical)	Escalation to department head

> **Clinical Tip:** Check the Follow-up worklist on the dashboard at the start of each day. Overdue tasks are prioritised by severity. High-risk patients with overdue visits appear at the top of the list.

Step 8 – Automatic Patient Management

This section demonstrates how GDES continuously manages the patient between visits.

Scenario: New Lab Results Arrive

Mr. Rahman returns for safety labs at 2 weeks. The results show:

Test	Baseline	2 Weeks	Change
Creatinine	1.6	1.55	Slight improvement
eGFR	48	50	Slight improvement
Potassium	4.2	4.3	Stable
Blood pressure	148/92	132/84	Improved (on ramipril)

Automatic Processing Chain

```
New Lab Results Entered
│
└─► Step 1: Trend Analysis
    ├─► eGFR: 48 → 50 (stable, no decline detected)
    ├─► Potassium: 4.2 → 4.3 (stable, no hyperkalemia)
    └─► Result: No critical alerts
│
└─► Step 2: Risk Recalculation
    ├─► Progression risk: Stable (no change)
    ├─► Follow-up interval: Unchanged (108 days)
    └─► Result: No change to plan
│
└─► Step 3: Clinical Reasoning
    ├─► Triggers: "Lab results updated"
    ├─► Engine: Reruns with new data
    ├─► Assessment: "Preserved eGFR on RAAS blockade"
    └─► Result: No change to differential
│
└─► Step 4: Follow-up Engine
    ├─► Monitors: All existing tasks
    ├─► Safety lab task: ✓ Completed
    └─► Result: Task marked complete, new tasks generated if needed
│
└─► Step 5: Dashboard Update
    ├─► Overview stats: Recalculated
    ├─► Worklist: Updated
    ├─► Patient trend chart: Updated with new data point
    └─► Result: Dashboard reflects latest status
│
└─► Step 6: Timeline Update
    ├─► Entry: "Lab results recorded"
    └─► Result: Timeline updated
│
└─► Step 7: Research Dataset
    ├─► Outcome engine: May trigger if sufficient data
    ├─► Research dataset: Updated with new values
    └─► Result: Research data always current
```

108-Day Follow-up Visit

Mr. Rahman returns for his scheduled follow-up visit. The results show:

Parameter	Baseline	108 Days	Change
Creatinine	1.6	1.7	Slight rise
eGFR	48	44	Decline of 4 mL/min
24h UTP	3.8 g	2.1 g	Improved (on RAAS blockade)
BP	148/92	128/80	Well controlled
Albumin	2.8	3.2	Improved
Weight	68 kg	65 kg	Edema resolved

Clinical Reasoning Update

- **eGFR slope:** −5.5 mL/min/year (computed from 3 data points)
- **Response:** Partial (proteinuria reduced >50%, but eGFR declining)
- **Assessment:** Consider immunosuppression (proteinuria still >1 g/day despite RAAS blockade, with evidence of progression)

Updated Follow-up Plan

Parameter	Before Visit	After Visit
Follow-up interval	108 days	90 days (increased risk)
Immunosuppression	Not started	Consider glucocorticoids
Next visit	—	90 days
Safety monitoring	Completed	New schedule for immunosuppression

---

Step 9 – Relapse Scenario

Mr. Rahman is started on glucocorticoids (prednisolone 0.8 mg/kg/day) after discussion of risks and benefits. He achieves partial remission over the next 6 months.

The Relapse

At 18 months after diagnosis, Mr. Rahman presents with:

- Recurrent bilateral leg edema (1 week)
- Foamy urine (reported by patient)
- Weight gain of 3 kg
- BP 136/84 mmHg (well controlled on ramipril)

Lab results:
Test	3 Months Ago	Today	Change
eGFR	46	38	**Decline**
24h UTP	0.8 g	3.5 g	**Relapse**

| Albumin | 3.6 | 2.9 | **Decline** |
| Creatinine | 1.6 | 2.0 | **Rise** |

Automatic Detection

```
New Lab Results Entered
│
└─► Step 1: Proteinuria Trend Analysis
    ├─► Previous: 3.8 → 2.1 → 0.8 (improving)
    ├─► Now: 3.5 (sudden rise)
    └─► Detection: Relapse likely
│
└─► Step 2: eGFR Trend Analysis
    ├─► Previous: Stable at 44-46
    ├─► Now: 38 (decline)
    └─► Detection: Acute kidney function decline
│
└─► Step 3: Clinical Reasoning
    ├─► Triggers: "Relapse detected"
    ├─► Engine: Full retrigger
    ├─► Disease phase: Active disease → Relapse
    ├─► Assessment: "IgAN relapse with AKI"
    └─► Updated risk: High
│
└─► Step 4: Follow-up Engine
    ├─► Interval: Shortened to 30 days
    ├─► New tasks:
    │   ├─► Urgent visit (within 1 week)
    │   ├─► Repeat labs (2 weeks)
    │   └─► Safety monitoring (glucocorticoid toxicity)
    └─► Escalation: Level 1 (clinician dashboard alert)
│
└─► Step 5: Disease Phase Update
    ├─► Previous: "Remission"
    ├─► New: "Relapse"
    └─► Timeline: "Relapse documented" entry
```

Clinical Assessment at Relapse Visit

At the urgent visit:

Encounter details:
- Type: *Unscheduled (relapse)*
- BP: *136/84 mmHg*
- Weight: *68 kg (+3 kg)*
- Edema: *2+*
- Clinician response: *Relapse*

Treatment adjustment:
- Glucocorticoids: Restarted at previous dose

- Consider: Second-line immunosuppression (discussed with patient)
- Plan: Recheck labs at 2 weeks, if inadequate response consider MMF or cyclophosphamide

Updated Follow-up Plan

Task	Due	Priority
Urgent labs (creatinine, eGFR, UTP)	2 weeks	High
Follow-up visit	30 days	High
Safety monitoring (glucocorticoid toxicity)	2 weeks	High
Full lab panel	30 days	Medium

Step 10 – Long-term Follow-up

GDES manages Mr. Rahman over several years. Here is the summary of his journey:

Time	eGFR	Proteinuria	Phase	Treatment
Baseline	48	3.8 g	Active	Ramipril started
3 months	44	2.1 g	Active	Prednisolone started
6 months	46	0.8 g	Partial remission	Prednisolone tapering
12 months	48	0.3 g	Complete remission	Prednisolone 5 mg
18 months	38	3.5 g	**Relapse**	Prednisolone increased
21 months	42	1.2 g	Improving	MMF added
24 months	44	0.4 g	Remission	MMF maintenance
36 months	42	0.3 g	Remission	Stable on MMF
48 months	40	0.2 g	Remission	Stable

Research Output

After 5 years of follow-up, Mr. Rahman's data contributes to the research dataset:

- **eGFR slope:** −1.6 mL/min/year (over 5 years)
- **Remission status:** Achieved complete remission after first relapse
- **Time to first remission:** 6 months
- **Time to relapse:** 18 months
- **Treatment episodes:** Ramipril, prednisolone (2 courses), MMF
- **Outcome:** Alive, no ESKD, stable kidney function

GDES's Long-term Value

After 5 years, GDES has maintained without any manual effort:

- A complete longitudinal clinical record
- An auditable trail of every decision and data change
- A continuously updated research dataset
- An automated follow-up schedule that prevented loss to follow-up
- A comprehensive outcome assessment ready for analysis

Automated Management

This chapter describes every automatic process in GDES, organised by trigger type.

Clinical Data Entry Automations

Trigger	Automation	Clinical Benefit
Patient registered	Audit log, timeline, research eligibility, follow-up initialisation	Complete record from first contact
Baseline saved	eGFR, BMI, BSA, KDIGO stage, CKD risk category computed	Risk stratification at enrolment
Lab result entered	Reference range checked, trend updated, critical alerts	Immediate recognition of abnormal values
Biopsy result entered	Disease classification, knowledge engine activated, protocol assigned	Guideline-based disease management
Prescription finalised	Treatment reconciliation, safety checks, monitoring tasks created	Medication management without extra work
Encounter saved	Disease phase assessed, visit recorded, next visit scheduled	Seamless visit-to-visit transitions

Monitoring and Detection Automations

Automation	Frequency	Description
eGFR slope computation	Every new lab result	Mixed-model slope estimation after 3+ results
AKI detection	Every new creatinine	KDIGO creatinine criteria (≥0.3 mg/dL in 48h or ≥1.5× baseline in 7 days)
CKD progression detection	Every new eGFR	KDIGO category change assessment
Proteinuria trend	Every new UTP/UPCR	Categorical change detection
Disease phase update	Every encounter	Active, remission, relapse, post-remission
Risk assessment	Every new clinical data	Progression risk, relapse risk, kidney survival

Scheduling Automations

Automation	Trigger	Description
Follow-up visit scheduled	Encounter saved, protocol assigned	Disease- and risk-appropriate interval

Automation	Trigger	Description
Lab schedule generated	Follow-up plan generated	Disease-specific lab panel timing
Medication monitoring	Active prescription	Drug-specific safety monitoring tasks
Safety monitoring	Immunosuppression	Infection surveillance, toxicity monitoring
Research visit	Study enrolment	Protocol-specific research visit schedule
Escalation trigger	Overdue task	Multi-level escalation chain

Continuous Automations

Automation	Description
Clinical reasoning	Reruns after every data change
Follow-up engine	Recomputes plan after every data change
Research dataset	Updated in real-time
Timeline	Every event automatically recorded
Audit log	Every change automatically recorded
Dashboard statistics	Recalculated periodically
Clinician worklist	Updated in real-time

Automated Follow-up Engine

The Follow-up Engine is the heart of GDES's automated patient management. It ensures no patient is lost to follow-up and that every patient receives the right care at the right time.

Architecture

```
┌──────────────┐
│  Patient     │
│  Data        │
└──────────────┘
       │
       ▼
┌──────────────┐     ┌──────────────────┐
│  Risk        │────►│  Protocol        │
│  Stratifier  │     │  Selector        │
└──────────────┘     └──────────────────┘
                              │
                              ▼
┌──────────────┐     ┌──────────────────┐
│  Task        │◄────│  Schedule        │
│  Generator   │     │  Calculator      │
└──────────────┘     └──────────────────┘
       │
       ▼
┌──────────────┐     ┌──────────────────┐
│  Worklist    │     │  Escalation      │
│  (Dashboard) │     │  Engine          │
└──────────────┘     └──────────────────┘
```

Risk Stratification

The risk stratifier evaluates 11 factors to determine follow-up intensity:

Factor	Low Risk	High Risk
eGFR	>60	<30
Proteinuria	<0.5 g/day	>3.5 g/day
Blood pressure	<130/80	>160/100
Histology (if available)	M0, S0, T0	M1, T1-2, crescents
Diabetes	No	Uncontrolled
Immunosuppression	None	High-dose steroids
Response to therapy	Complete remission	No response
Relapse history	First presentation	Multiple relapses

Factor	Low Risk	High Risk
Comorbidities	None	Multiple
Adherence	Good	Poor
Social factors	Stable	Unstable

Disease-Specific Protocols

GDES includes 10 disease-specific follow-up protocols:

Disease	Default Interval (Low Risk)	Default Interval (High Risk)
IgA Nephropathy	180 days	90 days
Minimal Change Disease	180 days	90 days
FSGS	90 days	60 days
Membranous Nephropathy	120 days	60 days
Lupus Nephritis	90 days	30 days
ANCA Vasculitis	90 days	30 days
Anti-GBM Disease	90 days	30 days
C3 Glomerulopathy	120 days	60 days
MPGN	120 days	60 days
Dense Deposit Disease	120 days	60 days

Task Types

The generator creates 7 types of follow-up tasks:

Task Type	Description	Example
Visit due	Clinical follow-up appointment	"Follow-up visit in 90 days"
Lab due	Laboratory investigation	"Serum creatinine, UTP"
Medication monitoring	Drug safety check	"CBC, LFT for MMF"
Safety monitoring	Adverse event surveillance	"Infection monitoring on steroids"
Research visit	Study protocol visit	"Study A: 6-month visit"
Referral due	Subspecialty referral	"Refer to ophthalmology for HCQ"
Patient contact	Phone call or reminder	"Remind patient of upcoming visit"

Escalation Chain

Missed tasks trigger a 4-level escalation:

```
Task Overdue
│
└─► Level 1 (1-14 days overdue)
    ├── Task marked "overdue" in worklist
    └── Visible on clinician dashboard
│
└─► Level 2 (15-30 days overdue)
    ├── Clinician notification sent
    └── Dashboard alert
│
└─► Level 3 (31-60 days overdue)
    ├── Clinic coordinator notified
    └── Patient contact initiated
│
└─► Level 4 (>60 days overdue)
    ├── Department head notified
    └── Case review triggered
```

> **Clinical Tip:** The escalation chain is your safety net. Even if you forget to schedule a follow-up, GDES will not let a high-risk patient fall through the cracks. Check the worklist daily to catch Level 1 escalations early.

How GDES Reduces Loss to Follow-up

1. **Automatic scheduling** — every patient has a follow-up plan from day one
2. **Risk-appropriate intervals** — high-risk patients are seen more frequently
3. **Centralised worklist** — all pending tasks visible in one place
4. **Multi-level escalation** — missed visits trigger increasingly urgent alerts
5. **Clinician dashboard** — overdue tasks are prominently displayed
6. **Timeline tracking** — complete visit history enables pattern recognition
7. **Outcome monitoring** — patients lost to follow-up are flagged in research analyses

---

Research Workflow

GDES is designed so that clinical care automatically contributes to research. This section explains how.

Automatic Cohort Inclusion

Every registered patient is automatically included in the research cohort. No separate enrolment process is required. If a patient withdraws consent, their data is flagged but retained for existing analyses.

Outcome Tracking

The outcome engine automatically computes:

Outcome	Definition	Data Required
Complete remission	Proteinuria <0.3 g/g, stable eGFR	Labs + clinical assessment
Partial remission	Proteinuria reduction >50%	Labs
No response	Does not meet remission criteria	Labs + clinical assessment
Relapse	Recurrence after remission	Labs + clinical assessment
ESKD	eGFR <15, dialysis, or transplant	Labs + events
Death	All-cause mortality	Events
Composite	ESKD, death, or sustained eGFR decline >40%	Labs + events

Research Data Export

GDES generates a denormalised one-row-per-patient research dataset.

To export:
1. From the Dashboard, click **"Export"** in the navigation sidebar.
2. Select format (CSV or Excel).
3. Optionally select identified (requires data manager role).
4. Click **"Export"**.

The Excel export includes a data dictionary sheet with column descriptions, types, and units.

Survival Analysis

From the Analytics module:

1. Select **"Survival Analysis"** from the Research page.
2. Choose outcome (ESKD, death, composite).
3. Optionally stratify by diagnosis, treatment, or other variables.
4. View Kaplan-Meier curves with confidence intervals.

5. View log-rank test results.
6. Export curves for publication.

Available analyses:
- Kaplan-Meier survival curves
- Nelson-Aalen cumulative hazard
- Log-rank test
- Incidence rate calculation
- Competing risks (cumulative incidence function)

Cox Regression

From the Analytics module:

1. Select **"Cox PH"** from the Research page.
2. Choose outcome and covariates.
3. View hazard ratios with confidence intervals.
4. View proportional hazards test.
5. Export results for publication.

Registry-Embedded Trials

GDES supports the full lifecycle of registry-embedded clinical trials:

1. **Study definition** — Define study, arms, eligibility criteria
2. **Eligibility screening** — GDES automatically identifies eligible patients
3. **Randomisation** — Stratified permuted-block randomisation
4. **Data collection** — Standard GDES data collection serves as trial data
5. **DSMB reporting** — Automatic safety tabulation by arm
6. **Analysis** — Standard GDES analytics applicable to trial data

Publication-Quality Datasets

The research dataset is designed to meet publication standards:

- Complete audit trail for every data point
- Standardised coding (diagnosis, medications, outcomes)
- Missing data explicitly coded
- Longitudinal structure supports time-to-event analysis
- De-identified by default (identified export gated to data managers)

Clinical Decision Support

GDES provides clinical decision support at every stage of patient management.

Evidence Grades

Each recommendation in GDES is assigned an evidence grade:

Grade	Meaning
Grade 1A	Strong recommendation, high-quality evidence
Grade 1B	Strong recommendation, moderate-quality evidence
Grade 1C	Strong recommendation, low-quality evidence
Grade 2A	Weak recommendation, high-quality evidence
Grade 2B	Weak recommendation, moderate-quality evidence
Grade 2C	Weak recommendation, low-quality evidence
No Grade	Consensus-based or expert opinion

Guideline References

Recommendations include specific references:

- KDIGO 2021 Glomerular Diseases Guideline
- ISN/RPS Lupus Nephritis Classification
- KDIGO Diabetes in CKD Guideline
- Local BIRDEM protocols
- Published trial results (TESTING, STOP-IgAN, etc.)

Reasoning Chain

The reasoning chain shows how GDES arrived at a conclusion:

```
 Example — IgA Nephropathy diagnosis:
1. Patient feature: Nephrotic syndrome (score: 15)
2. Patient feature: Microscopic hematuria (score: 10)
3. Patient feature: Reduced eGFR (score: 8)
4. Patient feature: Negative ANA (score: +5 toward primary GN)
5. Patient feature: Normal C3/C4 (score: +10 toward primary GN)
6. Knowledge rule: IgA dominant IF in biopsy (score: 60)
   Source: KDIGO 2021, Rule IG-01
   ------------------------------------------------
   Total score: 98/100 → IgA Nephropathy (98% confidence)
```

Confidence

Confidence scores reflect how well the patient's data matches the knowledge base rules:

Confidence	Meaning
>90%	Very likely — biopsy confirmation or strong syndromic match
70–90%	Likely — suggestive pattern with minor inconsistencies
40–70%	Possible — partial match, more data needed
<40%	Unlikely — weak match, alternative diagnoses more probable

Limitations

GDES decision support has limitations that clinicians should understand:

- **Knowledge base currency** — Rules reflect published evidence at the time of knowledge base update. Rare or newly described presentations may not be covered.
- **Data completeness** — Clinical reasoning is only as good as the data entered. Missing data reduces confidence and may produce less specific results.
- **Clinical judgment required** — GDES is a decision support tool, not a replacement for clinical judgment. All recommendations should be reviewed and contextualised by the treating clinician.
- **Local population considerations** — Knowledge base rules are derived from international evidence. Local disease patterns may differ.

Override Workflow

When a clinician disagrees with a GDES recommendation:

1. Document the override reason in the patient record
2. The override is recorded in the audit log
3. The clinical reasoning engine notes the override in future analyses
4. Override patterns can be reviewed for quality improvement

Practical Examples

This section provides additional case studies illustrating GDES use across different disease types.

Case 1 — Membranous Nephropathy

Presentation: 45-year-old female with nephrotic syndrome (UTP 6.2 g/day, albumin 2.2 g/dL), positive anti-PLA2R.

GDES workflow:
1. Registration → Baseline → Labs (positive anti-PLA2R)
2. Biopsy → PLA2R staining positive → MN confirmed
3. Clinical Reasoning → MN with high anti-PLA2R titre → risk assessed
4. Treatment → RAAS blockade + immunosuppression discussion
5. Follow-up → 60-day interval (high risk)

Monitoring:
- Anti-PLA2R titres serial monitoring
- Immunological remission (50% decline) detected at 6 months
- Clinical remission at 9 months

Case 2 — Lupus Nephritis

Presentation: 28-year-old female with malar rash, arthritis, proteinuria 2.8 g/day, positive ANA, anti-dsDNA, low C3.

GDES workflow:
1. Registration → Baseline → Labs (ANA+, dsDNA+, C3↓)
2. Biopsy → Class IV (diffuse proliferative) LN confirmed
3. Clinical Reasoning → Active LN with high activity score
4. Treatment → Methylprednisolone + MMF
5. Follow-up → 30-day interval (very high risk)

Monitoring:
- Complement recovery tracked
- Anti-dsDNA titres monitored
- Renal response at 6 months

Case 3 — ANCA Vasculitis

Presentation: 60-year-old male with rapidly progressive GN (creatinine 3.2 mg/dL, eGFR 18), positive MPO-ANCA, hematuria.

GDES workflow:
1. Registration → Baseline (urgent) → Labs (positive MPO-ANCA)
2. Biopsy → Pauci-immune crescentic GN
3. Clinical Reasoning → ANCA vasculitis with high-risk features

4. Treatment → Induction with cyclophosphamide + steroids
5. Follow-up → 30-day interval

Outcome:
- eGFR improved to 38 at 6 months
- ANCA negative at 12 months
- Maintenance azathioprine

Case 4 — Minimal Change Disease

Presentation: 22-year-old male with abrupt onset nephrotic syndrome (UTP 8.5 g/day, albumin 1.8 g/dL), normal eGFR, no hematuria.

GDES workflow:
1. Registration → Baseline
2. Biopsy → Minimal change disease (MCD)
3. Clinical Reasoning → MCD with high relapse risk identified
4. Treatment → Prednisolone 1 mg/kg/day
5. Follow-up → 90-day interval

Outcome:
- Complete remission at 4 weeks
- Steroid taper started
- Relapse at 12 months → second course

Case 5 — FSGS

Presentation: 38-year-old male with nephrotic syndrome (UTP 5.1 g/day), normal eGFR, no hematuria.

GDES workflow:
1. Registration → Baseline
2. Biopsy → FSGS, not otherwise specified
3. Clinical Reasoning → FSGS with high progression risk (Columbia classification)
4. Treatment → Prednisolone + RAAS blockade
5. Follow-up → 60-day interval

Case 6 — C3 Glomerulopathy

Presentation: 32-year-old female with hematuria, proteinuria 1.2 g/day, low C3.

GDES workflow:
1. Registration → Baseline → Labs (low C3, normal C4)
2. Biopsy → C3 dominant staining → C3 glomerulopathy
3. Clinical Reasoning → C3G differential (DDD vs C3GN)
4. Follow-up → 120-day interval

Case 7 — Kidney Transplant Recipient

Presentation: 42-year-old male with kidney transplant, new proteinuria 0.8 g/day.

GDES workflow:

1. Registration as transplant patient
2. Baseline assessment with transplant history
3. Labs → Screen for rejection, recurrence of primary disease
4. Biopsy if indicated
5. Follow-up → Individualised schedule

Appendices

Appendix A — Keyboard Shortcuts

Shortcut	Action
`/`	Focus search bar
`Ctrl+N`	New patient registration
`Alt+D`	Dashboard
`Alt+P`	Patient list
`Alt+A`	Analytics
`Alt+W`	Worklist

Appendix B — Dashboard Indicators

Indicator	Meaning
Live	System operating normally
Degraded	Partial functionality
Down	System unavailable
Badge (green)	Complete / Normal
Badge (yellow)	Pending / Warning
Badge (red)	Overdue / Critical

Appendix C — Common Clinical Codes

Code	Meaning
G1	eGFR ≥90
G2	eGFR 60–89
G3a	eGFR 45–59
G3b	eGFR 30–44
G4	eGFR 15–29
G5	eGFR <15
A1	Albuminuria <30 mg/g
A2	Albuminuria 30–300 mg/g

Code	Meaning
A3	Albuminuria >300 mg/g

Appendix D — Glossary

Term	Definition
AKI	Acute Kidney Injury
BSA	Body Surface Area
CKD	Chronic Kidney Disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration (eGFR formula)
DSMB	Data Safety Monitoring Board
eGFR	Estimated Glomerular Filtration Rate
ESKD	End-Stage Kidney Disease
GN	Glomerulonephritis
KDIGO	Kidney Disease: Improving Global Outcomes
MEST-C	Oxford classification scores for IgAN
MMF	Mycophenolate Mofetil
RAAS	Renin-Angiotensin-Aldosterone System
UPCR	Urine Protein-to-Creatinine Ratio
UTP	24-hour Urine Total Protein

Appendix E — Version History

Version	Date	Changes
5.1	July 2026	CDS reliability fixes (drug toxicity, treatment failure, relapse detection); human-readable disease names in CDS output; zero-confidence UI cleanup; CDS error logging; integration test suite (205 tests)
5.0	July 2026	Follow-up engine, clinical reasoning tab, audit log tab, GDES branding
4.0	April 2026	Clinical reasoning engine, knowledge base, decision support
3.0	January 2026	Biopsy module, central pathology review, research dataset export
2.0	October 2025	Prescription engine, treatment reconciliation, bilingual PDF
1.0	June 2025	Initial patient registry, baseline assessment, lab tracking

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GDES Version 5.1 — BIRDEM Nephrology, Dhaka, Bangladesh

For support: Contact the BIRDEM Nephrology IT Help Desk

This manual is intended for clinical users. No software engineering knowledge is required.